MIL-C-16751A (Wep)

26 JUNE 1963

SUPERSEDING
MIL-C-16751 (BuOrd)
13 March 1952

MILITARY SPECIFICATION

COVERS, CANVAS (FOR NAVAL ORDNANCE EQUIPMENT)

1. SCOPE

1.1 - Scope. This specification covers the manufacture of canvas covers, and applies to all types and classes of covers intended for protection of naval ordnance equipment.

1.2 - Classification. The canvas covers shall be one grade.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply:

SPECIFICATIONS

Federal

KK-L-271	Leather, Strap, Cattlehide, Vegetable-Tanned
V-T-276	Thread, Cotton
CCC-C-419 (Army & Numbered)	Cloth, Cotton, Duck, Unbleached, Plied Yarns
CCC-C-428	Cloth, Cotton, Duck, Fire, Water, Weather and Mildew Resistant
CCC-T-191	Textile Test Methods
T-C-571	Cord, Sash, Cotton, Braided
UU-T-81	Tags, Shipping and Stock

Military

MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-P-15159	Preservative Coating, Canvas (Navy)
MIL-G-16491 (Ships)	Grommets, Metallic

FSC 8305

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STANDARDS

Federal

FED-STD-751 Stitches, Seams and Stitching

PUBLICATIONS

Bureau of Naval Weapons

WR-12 Engineering Drawings, Associated Lists and Documents Referenced Therein

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 - Applicable drawings. Drawings applicable to the canvas covers shall be as specified in the contract or order. The covers shall be constructed in accordance with the applicable drawings, except where otherwise specified.

3.2 - Sample covers. When a sample cover is supplied in lieu of drawings, this specification shall take precedence wherever there is a difference between the sample cover and the specification. When a sample cover and drawings are supplied, the production cover shall be manufactured according to the drawings and this specification. The sample cover is supplied only for information and guidance.

3.3 - Material. The cover shall be made of number 8 treated cotton canvas, conforming to Specification CCC-C-428, Type II, Class 1. As an alternate, cotton canvas in accordance with Specification CCC-C-419, Type 1, having a preservative compound conforming to Specification MIL-P-15159 applied to both sides of the canvas, may be used. When the alternate method is used, the preservative compound shall be applied to the canvas prior to assembly of the cover. If for any reason it is

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necessary to vary the weight of the canvas it shall be so stated in the contract or order.

3.4 - Thread and Stitching. Thread shall be cotton thread Type 1A3 or 1B3, Ticket No. 12 as listed in Tables I or II of Specification V-T-276. No substitution of thread shall be made without permission of the contracting activity. The thread shall be black, blue, or as near the same color of the finished cover as possible. Stitching shall be in accordance with Specification FED-STD-751, Type 301 or 401. There shall be not less than 4-1/2 nor more than 5-1/2 stitches per inch. Machine needles larger than No. 22 shall not be used. Only lock stitching will be permitted; a chain stitch is not acceptable. When skipped or broken stitches occur, the stitches must be reinforced by parallel stitching for at least 2.0 inches on each side of the skipped or broken stitches. All stitching shall be waterproofed.

3.5 - Chafing Patches. All chafing patches shall be sewed to the inside of the cover. These patches shall be either canvas as described in 3.3, or leather, meeting the requirements of Specification KK-L-271, Selection A, Type A, Class 1, 5/64 inch minimum thickness.

3.6 - Lashing and Lacing. Lashings and lacings, as shown on the drawing (or sample cover if drawings are not available), shall be furnished by the contractor, and shall be braided sash-cord conforming to Specification T-C-571, Type 1, Class 1. All cord shall be treated with compound conforming to Specification MIL-P-15159.

3.7 - Metal Fittings. All grommets, ferrules and D-rings shall be of brass, unless otherwise specified on the drawings. Grommets shall also meet the requirements of Specification MIL-G-16491 (Ships). Ferrules shall be securely crimped. Metal fittings shall have no protruding sharp edges. D-rings shall be securely attached by means of cotton webbing conforming to Specification MIL-W-530, Type III.

3.8 - Hems and Seams. The hems and seams shall be as shown on the drawings; however, no hems are required on selvage edges, except as noted below.

For round grommets	2.0 inch hem
For oblong grommets	3.0 inch hem
For D-rings	3.0 inch hem

3.9 - Handles. Handles, when required, shall be cotton webbing, Type III, of Specification MIL-W-530.

3.10 - Deviation from Drawings, Sample Cover or Specifications. The contractor may at his option combine two or more of the parts on the drawings (or sample cover) or make one part into two or more parts to suit the width or cutting of the canvas, providing the original dimensions

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of the cover are maintained and providing that any new seams follow the pattern of the adjacent seams. Except for this, permission must be received from the contracting activity for any other deviations from the drawings or specifications.

3.11 - Color. Unless otherwise specified, the treated canvas shall be of the color specified in Specification CCC-C-428, Type II.

3.12 - Marking. Each cover shall be suitably marked in letters not less than one inch high with the markings as shown on the drawing or sample cover.

3.13 - Dimensions. No dimensions of the finished cover shall be less than that on the drawings or sample cover. However, the dimensions of the finished cover may be larger than the dimensions on the drawing or sample cover by the following amounts:

Dimensions on Drawings or Sample Covers	Plus Tolerances on Finished Production Cover
0 to 3.0 inches	3/8 inch
More than 3.0 to 12. inches	1/2 inch
More than 12.0 to 24.0 inches	3/4 inch
More than 24.0 inches	3 per cent

Allowances for shrinkage shall be in the directions indicated on the drawings or patterns. Otherwise, the dimensions of the finished cover should be as close as possible to the drawings or sample cover. Mating grommets, D-rings, or other approved closure devices shall match thruout the length of the lacing without noticeable bulging when laced.

3.14 - Workmanship. All details of manufacture shall be in accordance with first-class commercial practice covering this type of work. Covers shall be free from defects or blemishes affecting their appearance or serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 - Supplier's inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 - Sampling. One sample production cover shall be submitted to the activity contracting for the covers, or as directed in the contract, for inspection of material, details, workmanship, etc., so that it can be tried for fit on the equipment for which intended.

4.2 - Tests.

4.2.1 - Pre-manufacture tests. Prior to manufacture of covers, the canvas material shall have passed the tests called for in Specification CCC-C-428.

4.2.2 - Acceptance tests. The inspector shall request that covers be sample tested for strength and water tightness of seams. The number of each lot to be tested shall be determined by the inspector up to 100% test if deemed necessary.

4.2.2.1 - Water tightness test. Shape a pocket in the cover with the seam at the bottom. Fill the pocket with not less than one quart of water at ambient temperature and keep seam under water for one hour.

4.2.2.1.1 Leakage through the seam shall be cause for rejection.

4.2.2.2 - Strength test. The contractor shall submit sample seams, not less than two feet long, of each lot of thread used in the manufacture of the covers. The inspector shall ascertain that the sample seam is capable of withstanding at least a one hundred pound pull.

4.2.2.2.1 Failure of the seam at less than one hundred pound pull shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 - Packaging. Unless otherwise specified, each assembled cover shall be individually wrapped in Kraft paper or packaged in a cardboard carton, and securely tied or taped. Each assembled cover shall have an identification tag attached. Tags shall meet the requirements of Specification UU-T-81, Type A, Size No. 8.

5.2 - Packing. Unless otherwise specified, the packages shall be packed in substantial commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation at the lowest rate to the point of delivery. These boxes or crates shall be able to withstand storage, rehandling and reshipment without the necessity of repacking. All boxes shall fit the packages in a snug, tight-fitting manner. The boxes shall be suitably labeled to caution against the use of hooks.

5.3 - Marking.

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5.3.1 - Packages. Unless otherwise specified, packages shall be marked with the name of the Ordnance Material for which the cover is designed, contract number or order number and the name of the contractor.

5.3.2 - Shipping containers. Each container shall be marked "Canvas covers" and with the quantity contained therein, the name of the contractor, the number of the contract or order, and the gross weight.

6. NOTES.

6.1 - Ordering data. Procurement documents should specify the following:

(a) Title, Number and Date of this specification.
(b) Applicable drawing number.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implications or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R308

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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